STAT 6351: Assignment #6

Notes
This assignment is concerned with HFFD (Lecture 8).

Questions

1. (3M Stock: prices and transactions). The file `mmm9912-dtp.txt` in AFTS Ch 5 contains the transactions data of 3M stock in December 1999. There are three columns: day of the month, time of transaction in seconds from midnight, and transaction price. Transactions that occurred after 4:00 pm Eastern time are excluded.

   (a) Is there a diurnal pattern in 3M stock trading? Answer this question by constructing a time series of the number of trades in 5-minute intervals (script `hfntra.R`).

   (b) Use the price series to confirm the existence of a bid–ask bounce in intraday trading of 3M stock (script `hfchg.R`).

   (c) Use the data to construct an intraday 5-minute log return series (script `hfrtn.R`). Is the series serially correlated? Confirm this via the Ljung–Box test with lag 10.

   (d) There are 77 5-minute returns in a normal trading day. Some researchers suggest that the sum of squares of the intraday 5-minute returns can be used as a measure of daily volatility (the so-called *realized volatility*). Apply this approach to calculate the daily volatility of the log return of 3M stock in December 1999, and plot it.

2. (3M Stock: trading durations). The data file `mmm9912-adur.txt` in AFTS Ch 5 contains *adjusted* intraday trading durations of 3M stock in December 1999[1].

   (a) Is there a diurnal pattern in the adjusted duration series? Why?

   (b) Build a duration model for the adjusted series using exponential innovations. Check the fitted model.

   (c) Build a duration model for the adjusted series using Weibull innovations. Check the fitted model.

   (c) Build a duration model for the adjusted series using generalized gamma innovations. Check the fitted model.

   (d) Compare and comment on these three duration models.

[1]Details of this adjutment are as follows. There are 39 10-minute time intervals in a trading day. Let d_i be the average of all log durations for the ith 10-minute interval across all trading days in December 1999. Define an adjusted duration as $t_j/\exp(di)$, where j is in the ith 10-minute interval. Note that more sophisticated methods can be used to adjust the diurnal pattern of trading duration. Here we simply used a local average